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Class: TYBCA C
LAB 1 output screenshots

```
(venv) C:\Users\pranj\OneDrive\Desktop\Sem 5\ML lab\Lab_1>pip install numpy
Collecting numpy
  Downloading numpy-2.5.1-cp314-cp314-win_amd64.whl.metadata (6.6 kB)
  Downloading numpy-2.5.1-cp314-cp314-win_amd64.whl (12.6 MB)
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.6/12.6 MB 3.7 MB/s 0:00:03
Installing collected packages: numpy
Successfully installed numpy-2.5.1

(venv) C:\Users\pranj\OneDrive\Desktop\Sem 5\ML lab\Lab_1>pip install pandas
Collecting pandas
  Downloading pandas-3.0.5-cp314-cp314-win_amd64.whl.metadata (19 kB)
Requirement already satisfied: numpy>=2.3.3 in .\venv\Lib\site-packages (from pandas) (2.5.1)
Collecting python-dateutil>=2.8.2 (from pandas)
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting tzdata (from pandas)
  Downloading tzdata-2026.3-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
  Downloading pandas-3.0.5-cp314-cp314-win_amd64.whl (10.0 MB)
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10.0/10.0 MB 5.4 MB/s 0:00:01
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
  Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
  Downloading tzdata-2026.3-py2.py3-none-any.whl (348 kB)
Installing collected packages: tzdata, six, python-dateutil, pandas
Successfully installed pandas-3.0.5 python-dateutil-2.9.0.post0 six-1.17.0 tzdata-2026.3
```

```
(venv) C:\Users\pranj\OneDrive\Desktop\Sem 5\ML lab\Lab_1>pip install matplotlib
Collecting matplotlib
  Downloading matplotlib-3.11.1-cp314-cp314-win_amd64.whl.metadata (80 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.3.3-cp314-cp314-win_amd64.whl.metadata (5.5 kB)
Collecting cycler>=0.10 (from matplotlib)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib)
  Downloading fonttools-4.63.0-cp314-cp314-win_amd64.whl.metadata (121 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.5.0-cp314-cp314-win_amd64.whl.metadata (5.2 kB)
Requirement already satisfied: numpy>=1.25 in .\venv\Lib\site-packages (from matplotlib) (2.5.1)
Requirement already satisfied: packaging>=20.0 in .\venv\Lib\site-packages (from matplotlib) (26.3)
Collecting pillow>=9 (from matplotlib)
  Downloading pillow-12.3.0-cp314-cp314-win_amd64.whl.metadata (9.3 kB)
Collecting pyparsing>=3 (from matplotlib)
  Downloading pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: python-dateutil>=2.7 in .\venv\Lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in .\venv\Lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Downloading matplotlib-3.11.1-cp314-cp314-win_amd64.whl (9.5 MB)
 9.5/9.5 MB 3.9 MB/s 0:00:02
Downloading contourpy-1.3.3-cp314-cp314-win_amd64.whl (232 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.63.0-cp314-cp314-win_amd64.whl (2.3 MB)
 2.3/2.3 MB 7.3 MB/s 0:00:00
Downloading kiwisolver-1.5.0-cp314-cp314-win_amd64.whl (75 kB)
Downloading pillow-12.3.0-cp314-cp314-win_amd64.whl (7.2 MB)
 7.2/7.2 MB 7.9 MB/s 0:00:00
Downloading pyparsing-3.3.2-py3-none-any.whl (122 kB)
Installing collected packages: pyparsing, pillow, kiwisolver, fonttools, cycler, contourpy, matplotlib
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.63.0 kiwisolver-1.5.0 matplotlib-3.11.1 pillow-12.3.0 pyparsing-3.3.2
```

EXPLORER

ML LAB

.venv

.vscode

Lab_1

venv

assignment.py

Iris.csv

OUTLINE

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
(venv) C:\Users\pranj\OneDrive\Desktop\Sem 5\ML lab\Lab_1>python assignment.py
--- 1. First 5 Rows ---
  Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
0    1             5.1           3.5             1.4           0.2  Iris-setosa
1    2             4.9           3.0             1.4           0.2  Iris-setosa
2    3             4.7           3.2             1.3           0.2  Iris-setosa
3    4             4.6           3.1             1.5           0.2  Iris-setosa
4    5             5.0           3.6             1.4           0.2  Iris-setosa

--- 2. Random Sample (5 Rows) ---
  Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
10   11             5.4           3.7             1.5           0.2  Iris-setosa
25   26             5.0           3.0             1.6           0.2  Iris-setosa
55   56             5.7           2.8             4.5           1.3  Iris-versicolor
136  137             6.3           3.4             5.6           2.4  Iris-virginica
109  110             7.2           3.6             6.1           2.5  Iris-virginica

--- 3. Shape (Rows, Columns) ---
Rows: 150, Columns: 6

--- 4. Column Names ---
['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm', 'Species']

--- 5. Data Types ---
Id              int64
SepalLengthCm   float64
SepalWidthCm    float64
PetalLengthCm   float64
PetalWidthCm    float64
Species         str
dtype: object
```

EXPLORER...PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

ML LAB

- > .venv
- > .vscode
- Lab_1
 - > venv
- assignment.py
- Iris.csv

```
--- 6. Missing Values Count ---
Id                0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64

--- 7. Unique Species ---
<StringArray>
['Iris-setosa', 'Iris-versicolor', 'Iris-virginica']
Length: 3, dtype: str

--- 8. Species Value Counts ---
Species
Iris-setosa      50
Iris-versicolor  50
Iris-virginica   50
Name: count, dtype: int64

--- 9. Statistics for 'SepalWidthCm' ---
Mean: 3.0540
Median: 3.0000
Standard Deviation: 0.4336

--- 10. First 5 Rows (Without 'Id' Column) ---
   SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
0              5.1           3.5           1.4           0.2  Iris-setosa
1              4.9           3.0           1.4           0.2  Iris-setosa
2              4.7           3.2           1.3           0.2  Iris-setosa
3              4.6           3.1           1.5           0.2  Iris-setosa
4              5.0           3.6           1.4           0.2  Iris-setosa

(venv) C:\Users\pranj\OneDrive\Desktop\Sem 5\ML lab\Lab_1>
```